THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET A

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Average speed and the magnitude of average velocity are equal when an object: 1

(a) returns to start
(b) moves in a straight line in one direction
(c) is at rest
(d) moves in a circle

2. A child on a merry-go-round of radius 14 m completes one full revolution (π = 22/7). The distance travelled and the displacement are: 1

(a) 88 m and 28 m **(b)** 44 m and 14 m **(c)** 0 and 88 m **(d)** 88 m and 0

3. A body starts with velocity 0 m s^{-1} and accelerates uniformly at 4 m s^{-2} for 3 s. The distance travelled is: 1

(a) 18 m **(b)** 9 m **(c)** 12 m **(d)** 36 m

4. The weight of a 5 kg body is (g = 10 m s^{-2}): 1

(a) 5 N **(b)** 100 N **(c)** 50 N **(d)** 25 N

5. A fielder pulls his hands back while catching a fast ball to: 1

(a) increase the time of stopping and reduce the force
(b) increase the force
(c) increase the ball's mass
(d) decrease the ball's momentum to zero faster

6. Which of these is a pair of action–reaction forces? 1

(a) Paddle pushes water backwards; water pushes paddle forwards
(b) Two tug-of-war teams' pulls on a rope
(c) Gravity and friction on a car
(d) Weight of a book and the normal force on it

7. On a lever the load is 60 N at 20 cm from the fulcrum. To balance it with the effort arm 40 cm, the effort needed is: 1

(a) 15 N **(b)** 80 N **(c)** 30 N **(d)** 60 N

8. A machine does 600 J of work in 10 s. Its power is: 1

(a) 6000 W **(b)** 60 W **(c)** 120 W **(d)** 0.02 W

9. A constant force of 20 N displaces a body by 5 m in the direction of the force. The work done is: 1

(a) 100 J **(b)** 4 J **(c)** 25 J **(d)** 200 J

10. A source of frequency 1000 Hz produces waves of wavelength 0.34 m in a medium. The speed of the wave is: 1

(a) 340 m s^{-1} **(b)** 1000.34 m s^{-1} **(c)** 2941.18 m s^{-1} **(d)** 680 m s^{-1}

11. At a point in a medium, 10 density oscillations occur in 2 s. The frequency of the sound wave is: 1

(a) 10 Hz **(b)** 12 Hz **(c)** 5 Hz **(d)** 0.2 Hz

12. Sound cannot travel through: 1

(a) water **(b)** air **(c)** vacuum **(d)** steel

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Can the magnitude of displacement be greater than the distance travelled? Give reason. 2

14. Define one newton. 2

15. State Newton's third law of motion. 2

16. Write the expressions for kinetic and gravitational potential energies. 2

17. State the relation between speed, frequency and wavelength. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Draw a velocity–time graph for a body with constant acceleration and describe how acceleration is obtained. 3

19. Explain with two examples how Newton's second law helps in protecting from injury. 3

20. Explain why no work is done when you push hard against a wall. 3

21. Describe the bell jar experiment and state the conclusion. 3

22. Explain with the help of a graph the terms crest, trough and wavelength. 3

OR

Why is the minimum distance for an echo 17 m? 3

23. A man drives 200 km north in 3 h and then 200 km south in 2 h. Find the average speed and average velocity. 3

OR

A velocity–time graph is a horizontal line at 20 m s^{-1} from 0 to 6 s. Find the acceleration and displacement. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) How does a rocket lift off? 5
(ii) How is the same principle used by the Chandrayaan-3 lander?
(iii) Explain with the balloon activity.

OR

Case study: A 0.1 kg bullet leaves a 5 kg gun with a force of 2 N. 5
(i) What is the recoil force on the gun?
(ii) Find the bullet's acceleration.
(iii) Find the gun's acceleration.
(iv) Explain the difference.

25. (i) What is an echo and when is it heard? 5
(ii) Find the minimum distance of the reflector.
(iii) Distinguish echo and reverberation.
(iv) How are reverberations controlled in auditoriums?

OR

(i) How is sound produced? Describe an activity. 5
(ii) Show that sound needs a medium.
(iii) Why can astronauts not hear each other directly?

END OF QUESTION PAPER **Total: 50 Marks**